

Architecture

Lecture 1

Networks

- Architecture (Chapter 1)
- Application (Chapter 2 and 7)
- Protocols (Chapter 3 and 4)
- LANs (Chapter 5, 6.3)

No: Physical Layer, hardware, coding

Telephone networks design and Circuit switching

- 1 application(voice)
- long unicast connections/sessions(calls)
- almost constant traffic generation during the connection
- sparse(thinly dispersed) call arrival
- resource reserved for the entire call
- QoS (quality of service) guaranteed (once the connection is established)

Topologies

- fully interconnected network: every node connected to every other node)
- large number of links and the infrastructure is expensive
- centralized switch: all nodes connected to one central node
- small number of links, connects from any point to a central switch, very long wires
- two-level hierarchy: nodes connected to outside nodes that connect to interconnected inside ones.
Kind of/actually like what we are studying
- slightly larger number of links
- scalable infrastructure: many short connections, fewer long-distance connections
- Topologies with long distance (Multiple long-distance carriers)
- LATA: Local Access and Transport Area, LEC: local exchange carrier, IXC: interexchange carrier

Network of networks: ISP (internet service provider) -> IXP (internet exchange points) -> Tier 1
ISP (AT&T, Sprint, NTT, etc) consumer level to high level tier. Hosts connect to some ISP via
some access network that connects to these higher ones

Lines and trunks

- local loop line: often twisted-pair copper. Low capacity, analog
- trunks: often fiber. High capacity, digital
- high capacity: trunk lines carry more than 1 call simultaneously (thousands-millions)
- digital vs analog
- digital: regeneration, recovery vs analog: sensitive to noise

Lecture 2 Circuit & Packet switching (Practice Problems)

Telephone networks

- circuit switching
- lines and trunks
- multiplexing
- frequency division multiplexing (FDM)
- time division multiplexing (TDM)

Telephone Network Design

- telephone networks were engineered for voice
- service characteristics:
- constant bit rate
- call duration measured in minutes
- performance:
- throughput “ design variable
- delay “ firm deadline
- loss “ design variable

Computer Networks/Internet Design

- Packet Switching
- Idea: decompose message into packets (transmit the packets one by one)
- efficient for burst traffic
- packet switches/routers replace telephone routers (instead of setting up “circuits” for each call, route packets one by one) packets are buffered, FIFO
- no need of setting up a “circuit”™

-Statistical multiplexing

Internet Design

-was originally engineered for data and multiple applications (e.g. file transfer, email)

ABOVE

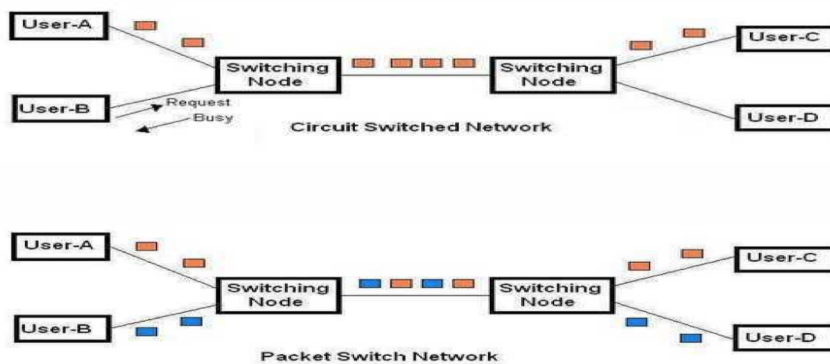
-bursty traffic generation

-short(frequency) connections

-packet switching, resource sharing, best effort(design variable)

-service characteristics:

-variable bit rate (burst)



-connection duration

measured in seconds

-performance:

-throughput “ design variable

-delay “ firm deadline

-loss “ often not ok (e.g. email)

Lecture 3(Resource sharing- trunks: Multiplexing)

Method 1-Circuit switching

-refers to creating an electrical circuit for the duration of each telephone call

-examples of telephone switches: telephone crossbar switch, telephone space division switch

Method 2- Frequency division multiplexing (FDM)

Method 3- Time Division Multiplexing (TDM)

Time and distance/ Time vs distance

-seconds, one hop = km

Propagation delay = distance / propagation speed

What are TDM Streams?

-Statistical (vs deterministic) TDM multiplexing

function of: traffic rate of other nodes, congestion

consequence: performance metrics (delay, throughput, packet loss probability) of an individual packet are: RANDOM VARIABLES

Synchronous vs. asynchronous(depending on queuing in routers)

-packets experience same delay vs different delays from source to destination

Connection-oriented vs connectionless (need packet ordering)

-packets arrive in sequence sent vs don't necessarily arrive in sequence sent

Reliable vs unreliable

Circuit switching vs Packet switching

Circuit switching: synchronous, connection-oriented, reliable

Datagram packet switching (without additional mechanism e.g. UDP): asynchronous, connectionless, unreliable

Datagram packet switching (with additional mechanisms e.g. TCP or in application):

-to get synchronous, add buffering before playout(e.g. streaming programs)

-to connection-oriented, add resequencing (e.g. TCP or in application)

-to get reliable, add retransmission (e.g. TCP or in application)

Why is the entire packet processed before it is forwarded?

[Header | Payload | Trailer]

-Error correction performed at each hop

-Packet regeneration (two steps: demodulation/remodulation, and error correction)

Cell phone networks

1G â€“ circuit-switched analog voice

2G â€“ circuit-switched digital voice & rudimentary data

3G â€“ circuit-switched digital voice & packet switched data

4G â€“ packet switched digital voice (VoIP) & packet switched data

Wired vs Wireless

-Wired: spatial multiplexing vs Wireless: broadcast channel interference

Cell structure

-Channel reuse: larger capacity, larger infrastructure cost, more uniform coverage

Cell phone networks

-MSCs (Mobile Switching Center) control end-to-end connection, channel assignment and handoff, and are connected to the PTN (Public telephone network) and internet

-To make calls: Access channel (contention based), Incoming calls: paging channel

GSM

-TDM: channels are assigned to multiple users

-BSC(Base station controller): control channel resource and handoff

-MSC(Mobile Switching Center): routes calls using the Visitor Location Register (local users) and the Home Location Register (last known location)

FDM + TDM: channels are assigned to multiple users

Downstream/upstream in different slots (half-duplex)

Cell phone Internet access

-2 merging families of standards (GSM, CDMA)

-FDM/TDM/CDMA used to define "channels"

-downstream (base station to mobile): packet switching, no contention! (no disagreement)

-upstream (mobile to base station): contention!, typically use a version of slotted ALOHA to reserve timeslots, also uses power and rate allocation

Cable TV Networks

Lecture 4 (Performance metrics)

Technology convergence

-Before vs Now

-Telephone networks were for telephone calls vs. telephone + internet + video

-Internet was for file transfer and email

-Cable TV networks for broadcast video vs. video + internet + telephone

-Cell phone networks(cellular networks) for telephone calls vs. telephone + internet + video

-Convergence:

-email, web browsing, streaming, telephone calls, video chat, gaming..

-over all networks

MORE INTERACTIVE LESS INTERACTIVE

Telephone calls Internet tv web browser email

Video conferencing Internet radio

BETTER PERFORMANCE WORSE PERFORMANCE

-different applications may have different Quality of Service (QoS) constraints

Packet switching performance metrics

-Packet delay, packet loss, throughput, efficiency

-Performance metrics are interdependent (e.g. no delay constraint = packet loss probability close to zero)

-End-to-end performance:

-Multi-hop

-Link (capacity, physical length, etc.)

-Node (processing speed, buffer)

-Topology

-Active end hosts

$d(\text{nodal}) = d(\text{processing is defined by the hardware}) + d(\text{queuing}) + d(\text{trans}) + d(\text{propagation})$

Delays

Packet = 512B (B = byte, b = bit)

Hop = 100 km

Transmission rate = 10 Mbps

-some examples might have multiple hops

Propagation delay example

-from the first bit transmitted to the first bit received

Propagation delay = distance / propagation speed

$$= 100\text{km} / (3 \cdot 10^8 \text{ m/s})$$

Transmission delay example

-from first bit pushed to last bit pushed

Transmission delay = packet size(in bits) / transmission rate

$$= (512\text{B})(8\text{bits/byte}) / (10 \cdot 10^6 \text{ bps})$$

Packet loss

What is the difference between queueing delay and (processing, transmission, and propagation delay?)

- Queueing delay is random

- processing, transmission, and propagation delay are deterministic

Throughput

- bits/second, calculated over a time window of many seconds

- what throughput will the user see in this web page download?

$\min(R(s), R(c))$

Efficiency

Data successfully delivered to the destination / Overall resource used

- includes overhead, idle time/frequency, lost packets, etc

Resource sharing!

Lecture 5 (Layering – the protocol stack)

- Internet = network of networks

- Provide a layering diagram

- Layers provide functionalities/services to the upper layers

- Some information exchange (e.g. failure reports)

- Not optimal

OSI Layering Model

OSI Layer 7: application

- implemented by hosts

- Applications: browsers, email programs, media players, etc.

- Protocols that support applications (type of messages, syntax): www:http; email: pop,smtp, imap; ftp, telnet, etc

OSI layer 4: transport

- end-to-end control

- TCP (transmission control protocol)

- segmentation

 - source and destination work together to decide pace (flow control)

 - retransmit dropped packets (reliable)

- packet ordering

- UDP (user datagram protocol)

OSI layer 3: network

- end-to-end packet forwarding

- Determine next router to send packet to

- Addressing

- routing

- Receive packet and determine next router to send packet to

- Queueing

- Routing

- Packet scheduling

OSI layer 2: link

- link scale services

- Channel access, who gets to transmit when

- framing

- error detection

- retransmission of corrupted packets

OSI layer 1: physical

- bits to analog signal

- modulation

- depends on transmission medium (e.g. twisted pair, coax, fiber, wireless)

- analog signal to bits

- synchronization

- sampling

- challenges: delay, attenuation, dispersion, noise

Layering, Layering + peering, packet format & layering

Internet Applications

Lecture 1

Application Layer

- Applications: Browsers, email programs, media players, etc
- Protocols that support applications (type of messages, syntax)
- www: http; email: pop, smtp, imap; ftp, telnet, etc.
- examples: email, file transfer, www, e-commerce, file sharing, streaming, gaming, voice over IP, video over IP

Application requirements

- Metrics: packet loss, delay, throughput
- Bounds: max/min, variations, firm/flexible

Application architecture: client-server

- client requests content: client must know server name or address
- server replies with content: server always on to listen for requests

Application architecture: peer-to-peer

- each machine acts as both a client and a server
- same task
- as a client: must have some way to find peer acting as a server that is on and has desired content
- as a server: must have some way of advertising availability and content
- scalable
- hybrid architectures

HyperText Transfer Protocol (HTTP)

- Purpose: request and transfer a webpage from a server to a user (client-server)
- Service characteristics
- pages requested and sent at random times
- connected only for duration of download
- Performance:
- loss: no okay
- delay: few seconds ok, but flexible
- throughput: higher is better, but flexible
- http: connection management
- connection initiation

- well-known ports (e.g. port 80)
- client (browser) ↔ server (webserver)
- connection access control (limited resources, TCP): server may close a connection
- server balancing: server may redirect a connection
- connection termination:
- non-persistent*: each file requires a separate connection
- first request main page, close connection
- if main page has images:
- for each image, open connection, request image, close connection
- workaround: open parallel connections, one for each image (but, server may have limited resources in terms of # of tcp connections)
- persistent:
- close connection only after client is done with server(or timeout)
- serial objects transfer
- TCP connection
- RTT ↔ round trip time (transmission, propagation)
- number of packets depends on file size
- timing of packets depends on layers below
- E-commerce over http
- purpose: purchase something on the www
- service characteristics:
- login using id and password
- select items
- checkout
- E-commerce requires “state”™
- “state”™ = information from previous communication (e.g. who you are, your basket, checkout status)
- problem: http doesn’t keep track of state
- solution: client and server application (above http) must jointly keep track of state
- client: cookie

-server: database

email

-Purpose: transfer a message from one user to another

-Service characteristics:

-messages sent at random times

-connected only for duration of upload or download

-Performance:

-loss “ not ok

-delay “ few minutes ok, but flexible

-throughput “ higher is better, but flexible

email: method & connection management

SMTP is used to transfer mail from the sender’s mail server to the recipient’s mail server; SMTP is also used to transfer mail from the sender’s user agent to the sender’s mail server. A mail access protocol, such as POP3, is used to transfer mail from the recipient’s mail server to the recipient’s user agent.

-well-known ports(e.g. 25, 110, 143)

-SMTP(simple mail transfer protocol)

-principal application-layer protocol for Internet electronic mail. It uses the reliable data transfer service of TCP to transfer mail from the sender’s mail server to the recipient’s mail server

mail access protocols

-POP(post office protocol):

- IMAP(internet mail access protocol) standards plus email format standard mean Alice and Bob don’t need to use the same email program

User agent: (client)

-dedicated program (e.g. outlook, elm, iphone mail)

-or webmail (e.g. uci webmail, gmail)

Mail server: (server)

-server that talks to a dedicated program (e.g. smtp.uci.edu, smtp.cox.net)

-server that runs webmail (all user agents and mail servers are at application layer)

Upload protocol:

SMTP(simple mail transfer protocol)

-open TCP connection, upload messages, close TCP connection

Download protocols:

POP = Post Office Protocol

-messages downloaded from mailbox on server to mailbox on non webmail client

IMAP = Internet Mail Access Protocol

-messages kept in (multiple) server mailboxes

-or moved to mailboxes on non webmail clients

email: location

-client-server or peer-to-peer?

two functions:

-upload/download (client-server)

-mail transfer: peer-to-peer (really server-to-server)

MAKE A CHART OF APPLICATION REQUIREMENTS EXAMPLE

Lecture 2

File sharing

-Purpose: find a file and transfer it from multiple other users

-Service characteristics:

-search for file

-identify which users have which pieces

-transfer pieces and put it together

-Performance:

-Loss: not okay

-Delay: very flexible, often hours

-Throughput: higher is better, but flexible

File sharing: search

-Location? (e.g. tracker)

-Client-server or peer-to-peer?

-client-server followed by peer-to-peer

-server: put peer in contact with peers

-peer-to-peer: file transfer

e.g. obtain list of peers from server then distribute to peers (file distribution with BitTorrent example diagram)

-Two functions:

-Search: usually client-server

-File transfer: peer-to-peer

File sharing: Method & Connection Management

-Method (e.g. BitTorrent)

-Connection Management

-often uses a large number of unregistered ports

-peers determined by protocol

-connections set up directly to peers

File sharing: action management

-BitTorrent:

-File splits into "chunks" ♦

-Client side: request missing chunks directly from other peers, via TCP

-Server side: list for and service requests for chunks you have, via TCP

-client algorithm determines which chunk to request first (e.g. rarest first)

-server algorithm determines rate (e.g. number of connections, max rate)

Multimedia

Streaming

-Purpose: 1 way transmission of audio or video

-Service characteristics:

-constant bit rate (unless compressed)

-duration: minutes to hours

-Performance:

-Loss: small amount ok

-delay: seconds, from once stream started

-throughput: fixed

Voice or Voice Over IP

- Purpose: 2 way interactive transmission of voice or video
- Service characteristics:
 - constant bit rate (unless compressed)
 - duration: minutes to hours
- Performance:
 - Loss: small amount ok
 - delay: a few tenths of a second
 - throughput: fixed
- Crude version: similar to streaming
- Better version: give priority to these packets over packets such as email or web browsing

Voice Over IP (VoIP)

- Real time application
- Multimedia streams
- Both directions

QoS Requirements of VoIP

- Packet loss
 - up to 20% is tolerated
- Reasons for packet loss: buffer overflow, link layer, delay
- UDP vs TCP: reliability (retransmission), delay, buffer starvation, delay variations
- End-to-end delay
 - Sum of all the delays (transmission, propagation, processing, queueing delays)
 - up to 400 ms tolerable
- Jitter
 - Packets generated periodically
 - delay variations at the receiver (e.g. queue conditions)

Best effort

- individual pkt end-to-end performance is random
- possibly large variables

-average may vary over time

Jitter-Countermeasure

-Timestamp: generation time

-delayed playout

-(most of the) packets arrive before playout time

-introduces delay

-packets after playout time are discarded

-delay is a random variable

-variations due to network conditions

-min delay with loss constraint

-fixed playout

- $t + q$

-generation + delay + max variations

-large variations = large delays

-adaptive playout

-talk spur = delay re-estimated

-fixed during talk spur

-recent measured delays used for estimation

Packet loss recovery

-TCP: retransmission, delay!

-Recovery: FEC, Interleaving, no additional RTT

FEC

-redundancy

+1 packet every N

N small: larger generation rate, better recovery

Delay: wait for the entire group

-low rate stream

-low-quality/low-bitrate stream appended

Interleaving

- samples are re-sequenced: adjacent samples assigned to different chunks
- packet loss mitigated: avoids gaps
- increased latency
- same bandwidth

Real Time Protocol (RTP)

- UDP
- RTP â€œ UDP â€œ IP
- RTP
- Independent RTP stream per source
- Video/audio payload + header
- RTP header fields (payload type, sequence number, timestamp, synchronization source identifier, miscellaneous fields)
- Payload type: encoding
- sequence number: re-sequencing, packet recovery
- timestamp: playout control
- synchronization source ID: identification

Supporting Multimedia Application

Supporting Multimedia

- Network dimensioning: enough bandwidth to support QoS
- Differentiated service: hierarchy of priorities
- Per-connection guarantees: end-to-end resource reservation

Dimensioning

- Avoid congestion: Links have enough bandwidth; No loss, small delay, small jitter etc.
- No changes to best-effort model
- End-to-end : Multiple ISP â€œ cooperation
- How much is enough

- â€œ Traffic demand
- â€œ Performance requirements
- â€œ End-to-end performance predic,on

Multiple service classes

- Multimedia/priority first, then the others
- priority per class and not per user/stream
- improved service
- avoids congestion
- Operations: packet marking, router processing, end-to-end
- issues
- too many prioritized streams or too much prioritized traffic
- congestion of low-priority traffic
- Solutions
- policing: traffic control (router), drop/delay packets
- fixed allocation: link level scheduling
- Efficiency?

Scheduling

- FIFO
- Priority queue
- Preemptive: service is interrupted
- Non-preemptive: service is not interrupted

Weighted round-robin

- round-robin: classes with non-empty queue sequentially served
- weighted fair queueing
- weight defines amount of time
- minimum fraction (due to empty queues)

The leaky-bucket

- limited injection of traffic in the buffers
- B tokens
- tokens assigned to incoming packets
- tokens generated with rate r
- policing
- average injection rate

-peak rate

-burst size

-multiple buckets

leaky-bucket + weighted round robin

FORMULA

DiffServ

-supports service differentiation

-edge functions: packet classification/marking, traffic conditioning

-core functions: per-hop behavior only function of the class

-traffic conditioning: pre-negotiated characterization, leaky bucket

Per-connection QoS

-end-to-end resource is pre-assigned

-QoS guarantees

-stream admission

-avoid unusable flow

-stream admission procedure

-setup signaling

-RSVP protocol

Connections

Traffic characterization by layer

CHAPTER 3!!

Whereas a transport-layer protocol provides logical communication between *processes* running on different hosts, a network-layer protocol provides logical communication between *hosts*.

The IP service model is a **best-effort delivery service and an unreliable service**

transport-layer multiplexing(collecting letters to send and giving them to delivery guy)

Extending host-to-host delivery to process-to-process delivery

-gathering data chunks at the source host from different sockets, encapsulating each data chunk with header information (that will later be used in demultiplexing) to create segments, and passing the segments to the network

demultiplexing: finding the right sockets to put the stuff through. job of delivering the data in a transport-layer segment to the correct socket is called

socket direction is determined by ALL the arguments in the tuple

UDP socket = (destination IP address and a destination port number)

-source port number = return address

TCP = source IP address, source port number, destination IP address, destination port number).

UDP

In fact, if the application developer chooses UDP instead of TCP, then the application is almost directly talking with IP.

Connectionless = no handshaking.

The network layer encapsulates the UDP segment into a datagram and sends the datagram to a name server. The DNS application at the querying host then waits for a reply to its query. If it doesn't receive a reply (possibly because the underlying network lost the query or the reply), either it tries sending the query to another name server, or it informs the invoking application that it can't get a reply.

Why UDP?

-no congestion control mechanism (more control but w/o congestion control there would be so much packet overflow and high loss rates)

-no connection establishment is faster

- -does not maintain connection state and does not track any of these parameters(needed to implement TCP's reliable data transfer service and to provide congestion control
- - *small packet header overhead*. The TCP segment has 20 bytes of header overhead in every segment, whereas UDP has only 8 bytes of overhead.

length field specifies the number of bytes in the UDP segment (header plus data)

UDP checksum provides for error detection

-provides error detection on an end to end basis but does not do anything to recover

sender will not send a new piece of data until it is sure that the receiver has correctly received the current packet : stop-and-wait protocols

Why layered architecture? Modularity, changes affecting only one layer at a time, standard interfaces

Every layer offers service to adjacent layers.

OSI layers

Different name data for different layers: message, segment, datagrams, frames, etc.

Encapsulation Layers:

Layer 7 = Application: HTTP, SMTP, FTP...

Layer 4: Transport: TCP, UDP

Layer 3: Network: Routing, Queuing, pck sched,

Layer 2: Link: Channel access, error detection, retransmission of corrupted Layer 1: Physical: depends on medium (e.g. modulation); synchronization, sampling, noise reduction, ...

Step in a HTTP request

First, we need to obtain the IP address of the webserver from the local DNS server.

- Second, we need to establish a TCP connection with the server. There are three messages in a TCP handshake: SYN, SYN-ACK and ACK.
- The third message (ACK) can also be used to send the HTTP request.
- Once the connection is ready, client sends a HTTP request and gets a response